

AIR QUALITY MONITORING AND ALERT SYSTEM

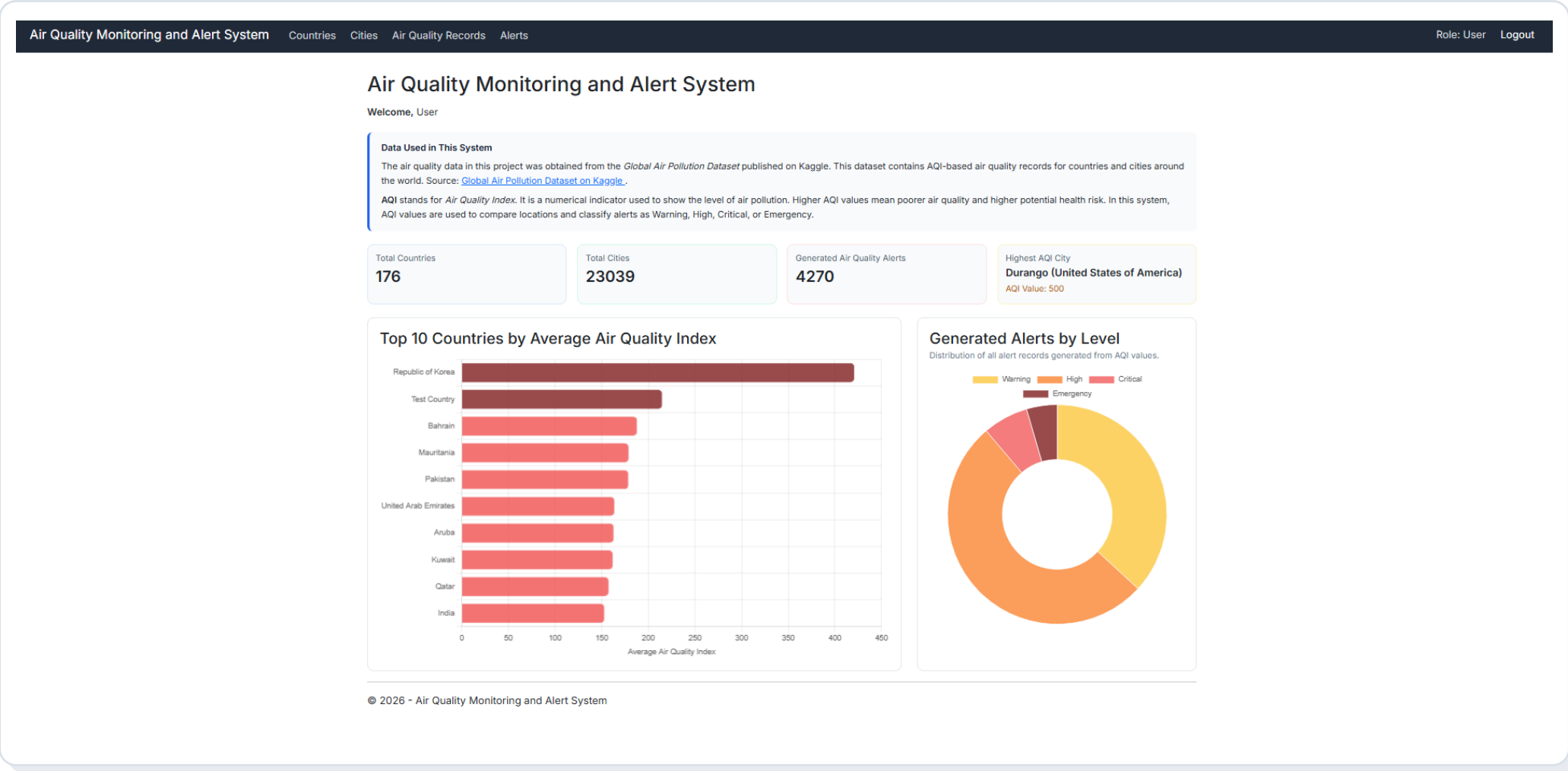
Visual Project Overview

Relational database project for monitoring AQI values, pollutant measurements, city-level air quality records, and automatically generated air-quality alerts.

CMPE 232 - Database Systems | Spring 2026 | TED University

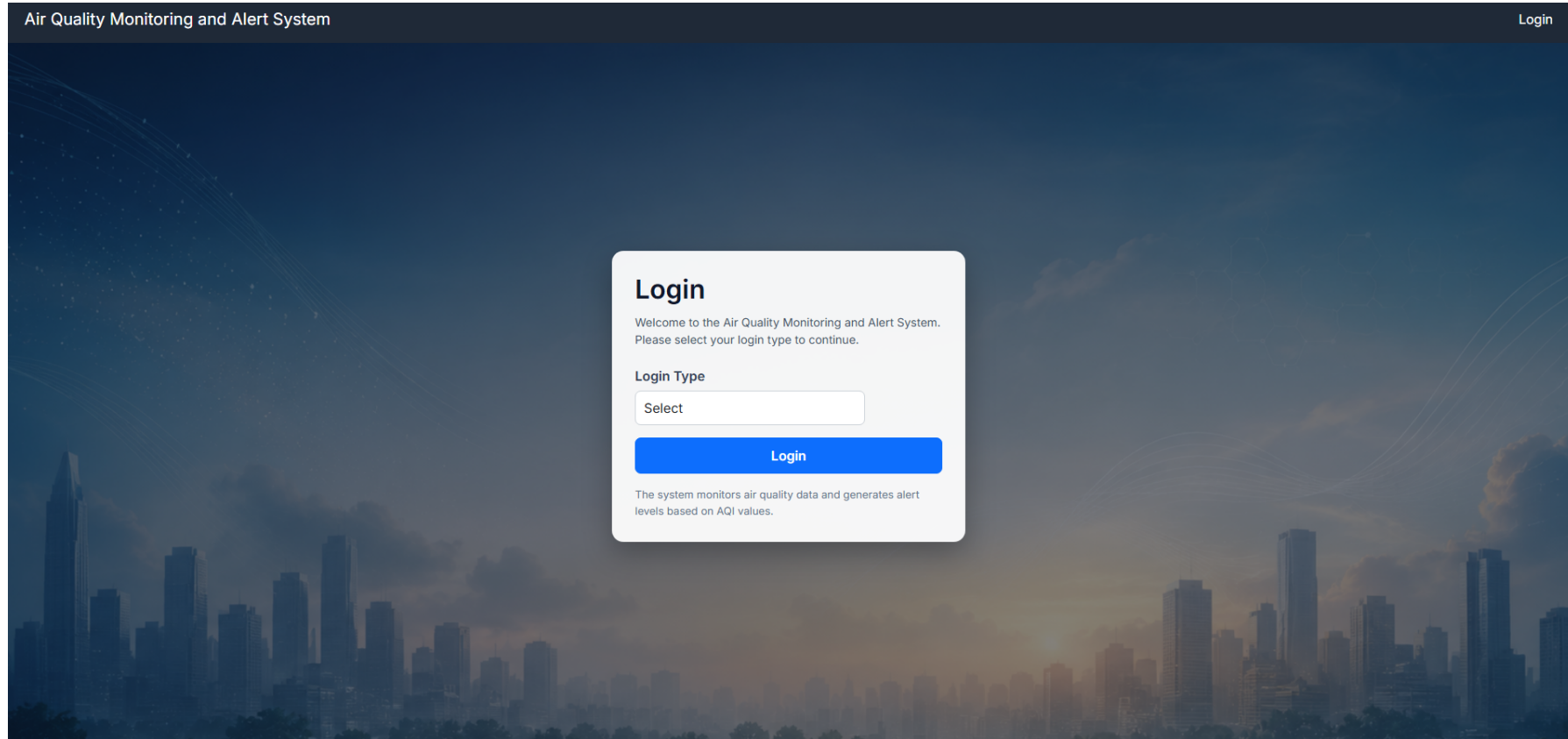
Air Quality Analytics Team (Sec02-Group16)

Beyza Yagmur Er | Semih Can Kasar | Guner Tugay Arisoy



User dashboard - global AQI overview and generated-alert distribution

Purpose - Store and organize country, city, pollutant and AQI data in a normalized relational database. - Detect unsafe air-quality conditions and generate alert records automatically.	Interface - User-facing pages provide search, filtering, pagination, sorting and AQI comparisons. - The dashboard summarizes dataset coverage, highest AQI information and alert distribution.
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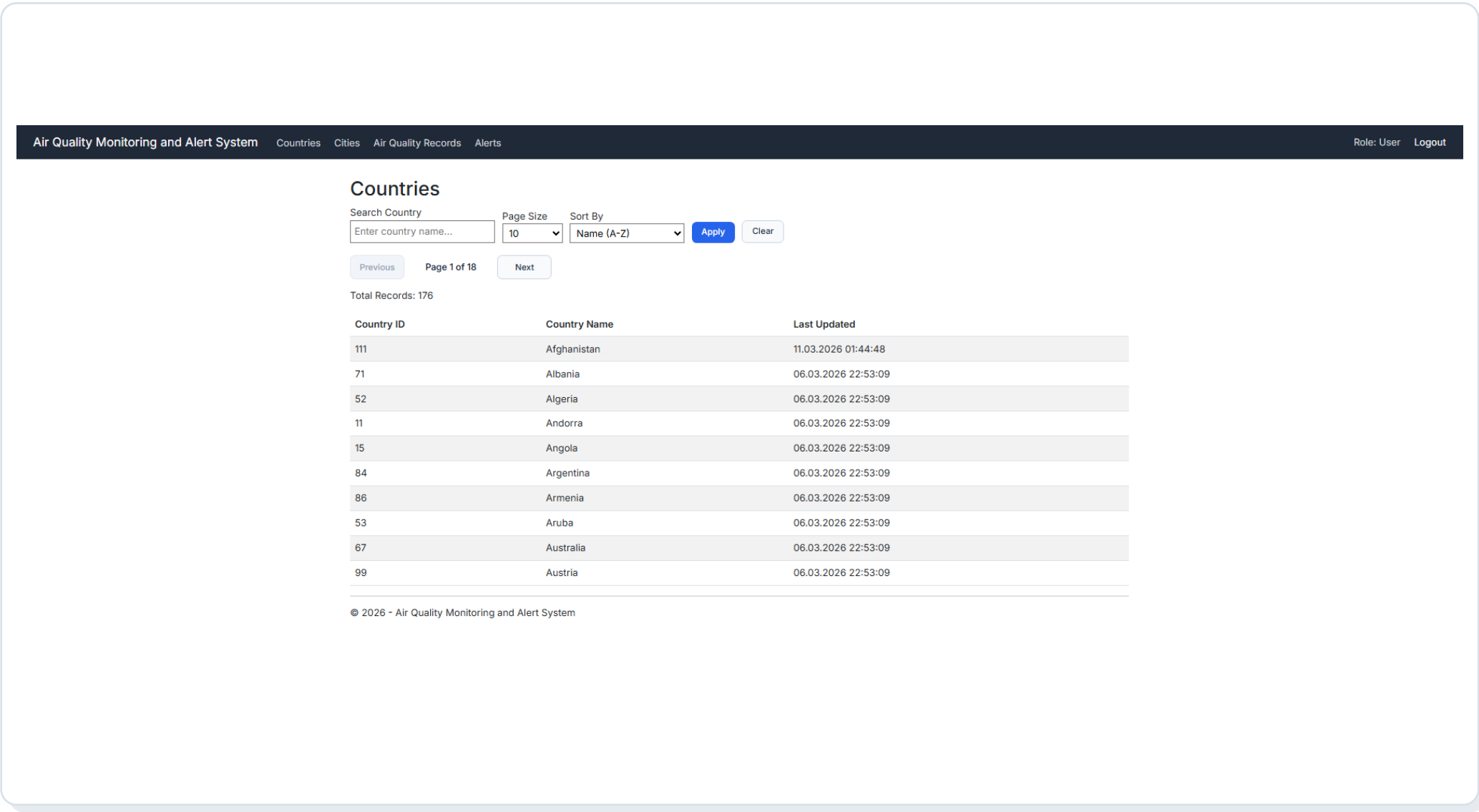
Login screen - login type selection

User Access

- Users can browse countries, cities, air-quality records and generated alerts.
- Data pages support search, filters, sorting, page-size selection and pagination.

Admin Access

- Selecting Admin exposes username and password fields before entering management pages.
- Administrative screens add create, edit, delete and record-management operations.



Countries page - search, sort and pagination for country records

What the screen demonstrates

- The Country entity is presented through a simple searchable table with Country ID, Country Name and Last Updated fields.

Air Quality Monitoring and Alert System

CountriesCitiesAir Quality RecordsAlerts

Role: UserLogout

Cities

Search City

Enter city name...

Page Size

10

Sort By

City Name (A-Z)

Apply

Clear

Previous

Page 1 of 2304

Next

Total Records: 23039

City ID	City Name	Country Name	Last Updated
18986	A Coruna	Spain	06.03.2026 22:53:30
20754	Aabenraa	Denmark	06.03.2026 22:53:30
7022	Aachen	Germany	06.03.2026 22:53:30
20755	Aalborg	Denmark	06.03.2026 22:53:30
3115	Aalborg	Netherlands	06.03.2026 22:53:30
7023	Aalen	Germany	06.03.2026 22:53:30
3116	Aalsmeer	Netherlands	06.03.2026 22:53:30
17912	Aalst	Belgium	06.03.2026 22:53:30
3117	Aalten	Netherlands	06.03.2026 22:53:30
17913	Aalter	Belgium	06.03.2026 22:53:30

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Cities page - city, country and last-updated information

Relational context

- Each city belongs to one country; the interface exposes the relationship directly through the Country Name column.

Air Quality Monitoring and Alert System

CountriesCitiesAir Quality RecordsAlerts

Role: UserLogout

Air Quality Records

Country

City

Pollutant

Page Size

Sort By

All Countries

All Cities

All Pollutants

10

Last Updated (Newest)

Apply

Clear

Previous

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Total Records: 23039

Country	City	Overall AQI Value	Overall AQI Category	CO AQI Value	CO AQI Category	Ozone AQI Value	Ozone AQI Category	NO2 AQI Value	NO2 AQI Category	PM2.5 AQI Value	PM2.5 AQI Category	Last Updated
Test Country	Test City 4	480	Hazardous	50	Good	90	Moderate	160	Unhealthy	480	Hazardous	16.05.2026 14:20
Test Country	Test City 3	180	Unhealthy	30	Good	80	Moderate	120	Unhealthy for Sensitive Groups	180	Unhealthy	14.05.2026 23:22
Test Country	Test City 2	130	Unhealthy for Sensitive Groups	20	Good	90	Moderate	110	Unhealthy for Sensitive Groups	130	Unhealthy for Sensitive Groups	14.05.2026 23:21
Test Country	Test City	70	Moderate	10	Good	40	Good	55	Moderate	70	Moderate	14.05.2026 23:20
Somalia	Bereda	101	Unhealthy for Sensitive Groups	1	Good	24	Good	0	Good	101	Unhealthy for Sensitive Groups	15.03.2026 17:36
Albania	Berat	91	Moderate	1	Good	47	Good	0	Good	91	Moderate	15.03.2026 17:21
Albania	Ballsh	55	Moderate	1	Good	49	Good	0	Good	56	Moderate	15.03.2026 16:56
Lao People's Democratic Republic	Champasak	47	Good	0	Good	22	Good	0	Good	47	Good	15.03.2026 16:42
Lao People's Democratic Republic	Luang Prabang	23	Good	0	Good	19	Good	0	Good	23	Good	15.03.2026 16:42
Lao People's Democratic Republic	Pakxe	29	Good	0	Good	19	Good	0	Good	29	Good	15.03.2026 16:42

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Air Quality Records - country/city/pollutant filters and AQI breakdown

Record structure

- The table presents Overall AQI together with CO, Ozone, NO2 and PM2.5 AQI values and their corresponding categories; records can be filtered and sorted.

Air Quality Monitoring and Alert System

CountriesCitiesAir Quality RecordsAlerts

Role: UserLogout

Alerts

Country

City

Alert Level

Page Size

All Countries

All Cities

All Levels

10

Apply

Clear

Previous

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Next

Total Alerts: 4263

Alert ID	Country	City	Pollutant	AQI Value	AQI Category	Alert Level	Alert Message	Last Updated
23772	Afghanistan	Anar Darreh	Overall AQI	126	Unhealthy for Sensitive Groups	Warning	Air quality warning in Anar Darreh. Causes: PM2.5 (AQI 126)	15/03/2026 16:43
23771	Afghanistan	Andarab	Overall AQI	104	Unhealthy for Sensitive Groups	Warning	Air quality warning in Andarab. Causes: PM2.5 (AQI 104)	15/03/2026 16:43
23770	Afghanistan	Asadabad	Overall AQI	131	Unhealthy for Sensitive Groups	Warning	Air quality warning in Asadabad. Causes: PM2.5 (AQI 131)	15/03/2026 16:43
23769	Afghanistan	Asmar	Overall AQI	151	Unhealthy	High	Air quality is unhealthy in Asmar. Causes: PM2.5 (AQI 151)	15/03/2026 16:43
23768	Afghanistan	Bagrami	Overall AQI	102	Unhealthy for Sensitive Groups	Warning	Air quality warning in Bagrami. Causes: PM2.5 (AQI 102)	15/03/2026 16:43
23767	Afghanistan	Cah Ab	Overall AQI	123	Unhealthy for Sensitive Groups	Warning	Air quality warning in Cah Ab. Causes: PM2.5 (AQI 123)	15/03/2026 16:43
23766	Afghanistan	Farah	Overall AQI	103	Unhealthy for Sensitive Groups	Warning	Air quality warning in Farah. Causes: PM2.5 (AQI 103)	15/03/2026 16:43
23765	Afghanistan	Geresk	Overall AQI	106	Unhealthy for Sensitive Groups	Warning	Air quality warning in Geresk. Causes: PM2.5 (AQI 106)	15/03/2026 16:43
23764	Afghanistan	Jurm	Overall AQI	128	Unhealthy for Sensitive Groups	Warning	Air quality warning in Jurm. Causes: PM2.5 (AQI 128)	15/03/2026 16:43
23763	Afghanistan	Kholm	Overall AQI	123	Unhealthy for Sensitive Groups	Warning	Air quality warning in Kholm. Causes: PM2.5 (AQI 123)	15/03/2026 16:43

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Alerts page - city, pollutant, AQI category, alert level and generated message

Warning

High

Critical

Emergency

Alert levels are generated from AQI severity; Good and Moderate conditions do not create alerts.

Air Quality Monitoring and Alert System Login

Login

Welcome to the Air Quality Monitoring and Alert System.
Please select your login type to continue.

Login Type

Admin

Username

Password

Login

The system monitors air quality data and generates alert levels based on AQI values.

Administration

- Administrative access is separated from standard user browsing and leads to data management capabilities

Air Quality Monitoring and Alert System

CountriesCitiesAir Quality RecordsAlerts

Role: AdminLogout

Countries

Search Country

Enter country name...

Page Size

10

Sort By

Name (A-Z)

Apply

Clear

Create New Country

Previous

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Next

Total Records: 176

Country ID	Country Name	Last Updated	Actions	
111	Afghanistan	11.03.2026 01:44:48	Edit	Delete
71	Albania	06.03.2026 22:53:09	Edit	Delete
52	Algeria	06.03.2026 22:53:09	Edit	Delete
11	Andorra	06.03.2026 22:53:09	Edit	Delete
15	Angola	06.03.2026 22:53:09	Edit	Delete
84	Argentina	06.03.2026 22:53:09	Edit	Delete
86	Armenia	06.03.2026 22:53:09	Edit	Delete
53	Aruba	06.03.2026 22:53:09	Edit	Delete
67	Australia	06.03.2026 22:53:09	Edit	Delete
99	Austria	06.03.2026 22:53:09	Edit	Delete

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Admin Countries page - Create New Country and row-level Edit/Delete actions

Management capability

- The admin interface adds explicit CRUD controls while retaining search, sort, pagination and last-updated visibility

Air Quality Monitoring and Alert System

CountriesCitiesAir Quality RecordsAlerts

Role: AdminLogout

Cities

Search City

Enter city name...

Page Size

10

Sort By

City Name (A-Z)

Apply

Clear

Create New City

Previous

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Total Records: 23039

City ID	City Name	Country Name	Last Updated	Actions	
18986	A Coruna	Spain	06.03.2026 22:53:30	Edit	Delete
20754	Aabenraa	Denmark	06.03.2026 22:53:30	Edit	Delete
7022	Aachen	Germany	06.03.2026 22:53:30	Edit	Delete
20755	Aalborg	Denmark	06.03.2026 22:53:30	Edit	Delete
3115	Aalborg	Netherlands	06.03.2026 22:53:30	Edit	Delete
7023	Aalen	Germany	06.03.2026 22:53:30	Edit	Delete
3116	Aalsmeer	Netherlands	06.03.2026 22:53:30	Edit	Delete
17912	Aalst	Belgium	06.03.2026 22:53:30	Edit	Delete
3117	Aalten	Netherlands	06.03.2026 22:53:30	Edit	Delete
17913	Aalter	Belgium	06.03.2026 22:53:30	Edit	Delete

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Admin_Cities.page - city management with associated country information

Referential management

- City management operates on records that are linked to Country through the relational schema and foreign-key structure

Air Quality Monitoring and Alert System

Countries

Cities

Air Quality Records

Alerts

Role: Admin

Logout

Air Quality Records

Country

All Countries

City

All Cities

Pollutant

All Pollutants

Page Size

10

Sort By

Last Updated (Newest)

Apply

Clear

Create New Record Set

Previous

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Total Records: 23039

Country	City	Overall AQI Value	Overall AQI Category	CO AQI Value	CO AQI Category	Ozone AQI Value	Ozone AQI Category	NO2 AQI Value	NO2 AQI Category	PM2.5 AQI Value	PM2.5 AQI Category	Last Updated	Actions
Test Country	Test City 4	480	Hazardous	50	Good	90	Moderate	160	Unhealthy	480	Hazardous	16.05.2026 14:20	Details
	Test City 3	180	Unhealthy	30	Good	80	Moderate	120	Unhealthy for Sensitive Groups	180	Unhealthy	14.05.2026 23:22	Details
	Test City 2	130	Unhealthy for Sensitive Groups	20	Good	90	Moderate	110	Unhealthy for Sensitive Groups	130	Unhealthy for Sensitive Groups	14.05.2026 23:21	Details
	Test City	70	Moderate	10	Good	40	Good	55	Moderate	70	Moderate	14.05.2026 23:20	Details
Somalia	Bereda	101	Unhealthy for Sensitive Groups	1	Good	24	Good	0	Good	101	Unhealthy for Sensitive Groups	15.03.2026 17:36	Details
Albania	Berat	91	Moderate	1	Good	47	Good	0	Good	91	Moderate	15.03.2026 17:21	Details
Albania	Ballsh	55	Moderate	1	Good	49	Good	0	Good	56	Moderate	15.03.2026 16:56	Details
Lao People's Democratic Republic	Champasak	47	Good	0	Good	22	Good	0	Good	47	Good	15.03.2026 16:42	Details

Admin Air Quality Records - Create New Record Set and Details actions

AQI record management

- Administrators can create new record sets and inspect record details while using the same country, city, pollutant and sorting filters available to users.

Relational Design

- Normalized to Third Normal Form (3NF) to reduce redundancy and maintain data integrity.
- Core entities: Country, City, Pollutant, GasPollutant, ParticulatePollutant, AQICategory, AirQualityRecord, Alert and AlertNote.
- RawAirQualityImport is used as a staging structure before data is transformed into normalized tables.
- Primary keys, foreign keys, unique constraints and a non-negative AQI CHECK constraint enforce consistency.

Database Automation

- LastUpdated triggers track modifications across database entities.
- Overall AQI can be recalculated from pollutant records for affected cities.
- A database trigger creates or updates alerts for Warning, High, Critical and Emergency conditions.

Team and Work Distribution

SQL Part	Semih Can Kasar, Beyza Yagmur Er
UI Design	Guner Tugay Arisoy
Relational Modeling	All team members
Report & Writing	Beyza Yagmur Er, Semih Can Kasar

Project Scope

- Monitors AQI information across countries and cities.
- Stores pollutant-specific records for CO, Ozone, NO2 and PM2.5.
- Classifies AQI values and generates alerts for unsafe conditions.
- Provides role-based user and administrator interfaces for viewing and managing data.